

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

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L6.S1.A

1 Random Variables (RVs)

1.1 Motivation and Concept

A random variable (RV) is a function that maps outcomes of a random experiment (sample space) to numerical values.

The distribution of a random variable can be visualized using a bar diagram.

The x-axis represents the values that the random variable can take.

The height of the bar at value a_i represents:

$$\Pr[X = a_i]$$

Each probability is computed by finding the probability of the corresponding event in the sample space.

2 Types of Random Variables

2.1 Discrete Random Variables

A discrete random variable has the following properties:

- Countable support
- Defined using a probability mass function (PMF)
- Probabilities are assigned to single values
- Each possible value has strictly positive probability

2.2 Continuous Random Variables

A continuous random variable has the following properties:

- Uncountable support
- Defined using a probability density function (PDF)

- Probabilities are assigned to intervals of values
- Each specific value has zero probability

3 Example: Random Variable from Coin Tossing

Example 1: Tossing 3 Fair Coins

Experiment: Toss 3 fair coins. Let Y denote the number of heads observed.

Possible values of Y :

$$Y \in \{0, 1, 2, 3\}$$

Probability Computation:

$$P(Y = 0) = P(t, t, t) = \frac{1}{8}$$

$$P(Y = 1) = P(t, t, h), (t, h, t), (h, t, t) = \frac{3}{8}$$

$$P(Y = 2) = P(h, h, t), (h, t, h), (t, h, h) = \frac{3}{8}$$

$$P(Y = 3) = P(h, h, h) = \frac{1}{8}$$

Verification of Total Probability:

Since Y must take one of the values 0 through 3,

$$1 = P\left(\bigcup_{i=0}^3 \{Y = i\}\right) = \sum_{i=0}^3 P(Y = i)$$

4 Probability Mass Function (PMF)

4.1 Definition

A random variable that can take at most a countable number of values is called discrete.

Let X be a discrete random variable with range:

$$R_X = \{x_1, x_2, x_3, \dots\}$$

where the range is finite or countably infinite.

The function

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the Probability Mass Function (PMF) of X .

4.2 PMF Property

Since X must take one of the values x_k , we must have:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

5 PMF Example: Given Functional Form

Example

The PMF of a random variable X is given by:

$$p(i) = c \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Tasks:

1. Find $P(X = 0)$
2. Find $P(X > 2)$

Step 1: Determine Constant c

Since probabilities must sum to 1:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the known series:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

We get:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Step 2: Compute $P(X = 0)$

$$P(X = 0) = e^{-\lambda} \frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $P(X > 2)$

$$\begin{aligned}P(X > 2) &= 1 - P(X \leq 2) \\&= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\&= 1 - \left(e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda} \right)\end{aligned}$$

6 Bayes' Theorem (Recap)

6.1 Conditional Probability Identity

Using:

$$\Pr(A|B) = \Pr(B|A) \Pr(A)$$

We obtain:

$$\Pr(B_i|A) = \frac{\Pr(A|B_i) \Pr(B_i)}{\sum_{j=1}^n \Pr(A|B_j) \Pr(B_j)}$$

This is known as Bayes' Formula (Proposition 3.1).

6.2 Terminology

$\Pr(B_i)$: A priori probability (probability formed from self-evident or presupposed models)

$\Pr(B_i|A)$: Posterior probability (probability computed after observing event A)

7 Bayes' Theorem – Example 1: Auditorium with 30 Rows

Problem Setup

An auditorium has 30 rows.

Row 1 has 11 seats Row 2 has 12 seats Row 3 has 13 seats

$\vdots$

Row 30 has 40 seats

A door prize is awarded by:

1. Randomly selecting a row (each row equally likely)
2. Randomly selecting a seat within that row

Task 1: Probability that Seat 15 is Selected Given Row 20

$$P(S_{15}|R_{20}) = \frac{1}{30}$$

Task 2: Probability that Row 20 was Selected Given Seat 15

Using Bayes' rule:

$$P(R_{20}|S_{15}) = \frac{P(S_{15}|R_{20})P(R_{20})}{P(S_{15})}$$

Where:

$$P(S_{15}) = \sum_{k=5}^{30} P(S_{15}|R_k)P(R_k)$$

Numerical evaluation (as shown in lecture):

$$P(S_{15}) = 0.0342$$

Final result:

$$P(R_{20}|S_{15}) = \frac{\frac{1}{30} \cdot \frac{1}{30}}{0.0342} = 0.0325$$

8 Bayes' Theorem – Example 2: Communication System (Receiver RX)

A communication system sends binary data (0 or 1).

The receiver (RX) may make mistakes:

- A 0 may be detected as 1
- A 1 may be detected as 0

The system is described using a set of conditional probabilities, and Bayes' theorem is used to compute posterior probabilities of transmitted symbols given received symbols.